Proof

If $\vec{n}_1$ and $\vec{n}_2$ are the vectors normal to the two given planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ respectively. Then, $\vec{n}_1=a_1\hat{i}+b_1\hat{j}+c_1\hat{k}$ and $\vec{n}_2=a_2\hat{i}+b_2\hat{j}+c_2\hat{k}$

Therefore, using equation (1) in theorem 6.18 the acute angle θ between the planes is given by

$$\theta=\cos^{-1}\left(\frac{|a_1a_2+b_1b_2+c_1c_2|}{\sqrt{a_1^{\,2}+b_1^{\,2}+c_1^{\,2}}\sqrt{a_2^{\,2}+b_2^{\,2}+c_2^{\,2}}}\right)$$

Remark

(i) The planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ are perpendicular if $a_1a_2+b_1b_2+c_1c_2=0$

(ii) The planes $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$ are parallel if $\frac{a_1}{a_2}=\frac{b_1}{b_2}=\frac{c_1}{c_2}$

(iii) Equation of a plane parallel to the plane $ax+by+cz=p$ is $ax+by+cz=k$, $k\in\mathbb{R}$

Example 6.47

Find the acute angle between the planes $\vec{r}\cdot\left(2\hat{i}+2\hat{j}+2\hat{k}\right)=11$ and $4x-2y+2z=15$.

Solution

The normal vectors of the two given planes $\vec{r}\cdot\left(2\hat{i}+2\hat{j}+2\hat{k}\right)=11$ and $4x-2y+2z=15$ are $\vec{n}_1=2\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{n}_2=4\hat{i}-2\hat{j}+2\hat{k}$ respectively.

If θ is the acute angle between the planes, then we have

$$\theta=\cos^{-1}\left(\frac{|\vec{n}_1\cdot\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|}\right)=\cos^{-1}\left(\frac{\left|\left(2\hat{i}+2\hat{j}+2\hat{k}\right)\cdot\left(4\hat{i}-2\hat{j}+2\hat{k}\right)\right|}{\left|2\hat{i}+2\hat{j}+2\hat{k}\right|\left|4\hat{i}-2\hat{j}+2\hat{k}\right|}\right)=\cos^{-1}\left(\frac{\sqrt{2}}{3}\right)$$

6.8.11 Angle between a line and a plane

We know that the angle between a line and a plane is the complement of the angle between the normal to the plane and the line

Let $\vec{r}=\vec{a}+t\vec{b}$ be the equation of the line and $\vec{r}\cdot\vec{n}=p$ be the equation of the plane. We know that $\vec{b}$ is parallel to the given line and $\vec{n}$ is normal to the given plane. If θ is the acute angle between the line and the plane, then the acute angle between $\vec{n}$ and $\vec{b}$ is $\left(\frac{\pi}{2}-\theta\right)$. Therefore,

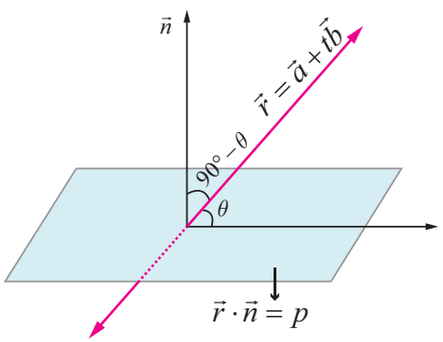


Fig. 6.31

$$\cos\left(\frac{\pi}{2}-\theta\right)=\sin\theta=\frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}$$

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